Unit II: The Demand for Health and Health Care

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Health Economics

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References

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Unit II: Demand for health and health care

- Medical care and health care
- Is health care a Commodity?
- Is health a Commodity?
- Concepts of Utility and Demand
- Health and Utility
- Medical care and utility
- Demand for medical Services
- Economic and non-economic factors affecting demand for health care

Unit II: Demand for health and health care

- Fuzzy Demand Curve
- Elasticity of demand for medical care
- Factors in estimating elasticity in demand for health care
- Empirical Findings
- Economic perspective of health
- Grossman Model
- Anderson and Newman Model

Demand for Medical Care

- What are the factors (economic and non-economic) that determine the demand for medical care?

- How health is similar and different to other commodity ?

- What are the framework on demand for medical care?

Drawing Graphs

- ❑ Draw a Utility Curve for health
- ❑ Draw a Production Function of health
- ❑ Draw a Demand Curve for health care

Health

Health is not everything but without health, life is nothing

1. Health is highly valued Assets

2. Health is pre-requisite for other activities

Health and its Determinants

Health = f(age, sex, education, income, occupation, alcohol consumption, smoking, diet, exercise, medical care, genetic factors, environment, etc)

Health = f(Medical Care)

Medical Care is Proximate Determinant of Health

What is Medical Care?

Health = f(Medical Care)

Medical care is composed of **goods and services** that maintain, improve or restore a person's physical or mental well-being

Ex Medical Goods??

Ex Medical Services??

Medical Services exhibit 4 I’s

1. Intangible

2. Inseparable

3. Inventory

4. Inconsistency

Health Services

- ❑Use of health services is **unpleasant**
- ❑At what phase of life you need healthcare more?
- ❑Is health care a necessity?

Health Care and Medical Care

Health care: Anything that contribute to producing better health such as **nutritious food, clean air, <u>medical care</u> and exercise** (Johnson-Lans 2006)

Medical care: is composed of **<u>goods and services</u>** that maintain, improve or restore a person's physical or mental well-being

Is Health Care a Commodity?

Three characteristics of health care

1. **Uncertainty** with respect to need for and effectiveness of health care
2. Information **Asymmetries** between provides and patients
3. Externalities

Compare Demand for Medicare care and Demand for Food

Medical service is associated with

a) Risk to death
b) Considerable Risk to impairment
c) Potential reduction of earnings

Food is a necessity

Deprivation from food can be guaranteed with income but that is not possible in case of medical care

The behavior expected of a seller of medical care is different from business man

Is Health a Commodity?

- What are the features of a commodity?
- Health is not **tradable**
- Improvement in health **can not be purchased** directly

Is Medical care a Commodity?

- Nature of Demand
- Product uncertainty

- Problem information

- Supply Condition
  - Large role of Nonprofit Firms
  - Restrictions on competitions

- Govt Subsidy and Public Provision
- Pricing Practices

- Externalities

1.Nature of Demand : Irregular and Unpredictable

Medical care has a special place in Economic Analyses

Nature of Demand: Demand for health care is irregular and unpredictable in nature from individual perspective and also a health care firm

Not steady in nature like food or clothing
What is nature of demand for food?

Afford satisfaction only in case of illness

Demand for medical services has risk of death and impairment

Arrow: Uncertainty and Welfare Economics of Medical Care

2. Product uncertainty

- Recovery from disease is as unpredictable as its incidence
- Consumer are uncertain in their health status and need for health care in future
- In many cases consumer often do not know the expected outcome of various treatment without physician's advice and in many cases physician can not predict the outcome of interest with certainty

3.Asymmetry of information

- Consumer may not know which physicians or hospital are good, capable and competent

- Sometime information is unavailable to all parties concerned.

- Ex: Neither Gynecologist nor their patients may recognize the early sign of cervical cancer without Pap Smeers.

- At other times, the information is known to some parties but not all.

4. Supply Condition

Behaviour of a seller of medical care different from a business men

Consumer can not taste the product before consuming it

Open price competition are absent

5.Large role of Nonprofit Firms

- Economist often assumes that firm maximises profit.

- But, many health care providers including many hospitals , insures and nursing home have non-profit status

6.Restrictions on competitions

- License requirement for providers, restriction on provider advertising and standard of ethical behavior

- Regulation to promote quality or to curb costs also reduce the freedom of choice of providers and may influence competition

7. Govt Subsidy and Public Provision

- In most countries, Govt plays a major role in the provision of financing health care services

- Health and healthcare market presents an unusual features that makes a unique discipline

8.Pricing Practices

- Price discrimination by Income
- Higher fees for prepayment: Premium insurance

9. Externalities

- Externalities is external benefit/cost
- Does flu has external cost?
- Does COVID-19 has external cost?

Economic Concepts

1. Marginal utility and total utility
2. Law of Diminishing Marginal Utility
3. Law of Demand
4. Elasticity of Demand
5. Law of Diminishing Marginal Productivity

Medical Care and Utility

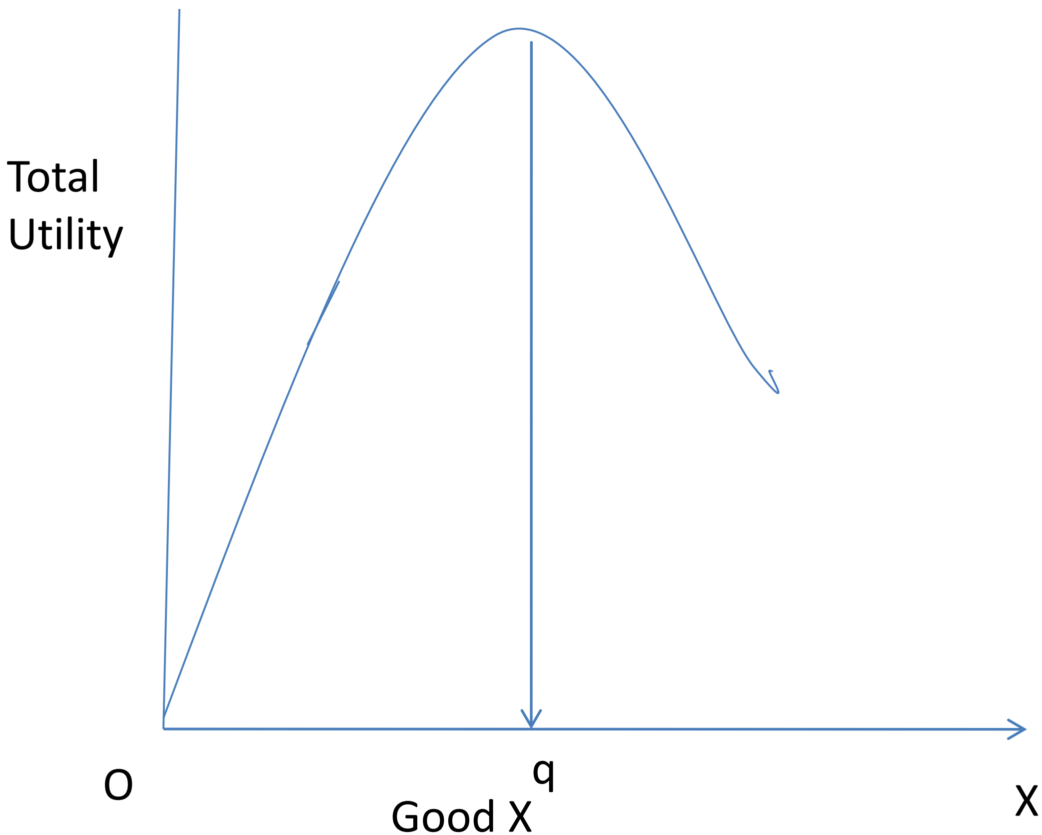
- Health yields utility to the consumer

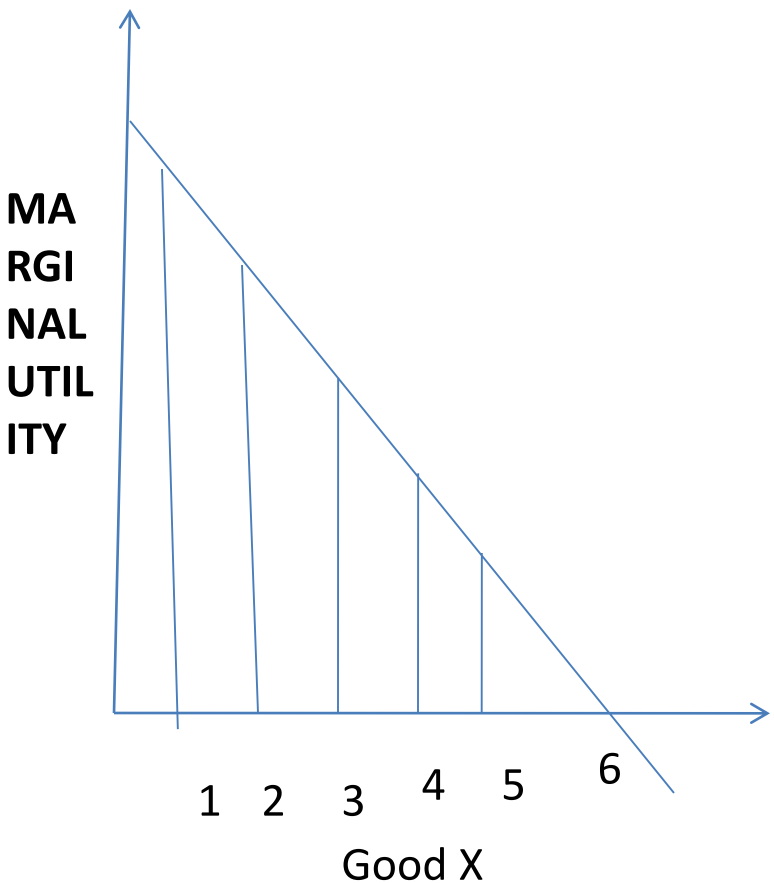
→ Subject to **law of diminishing marginal utility**

- Medical care is an input in producing health

→ Subject to **law of diminishing marginal productivity**

Marginal Utility and Total Utility

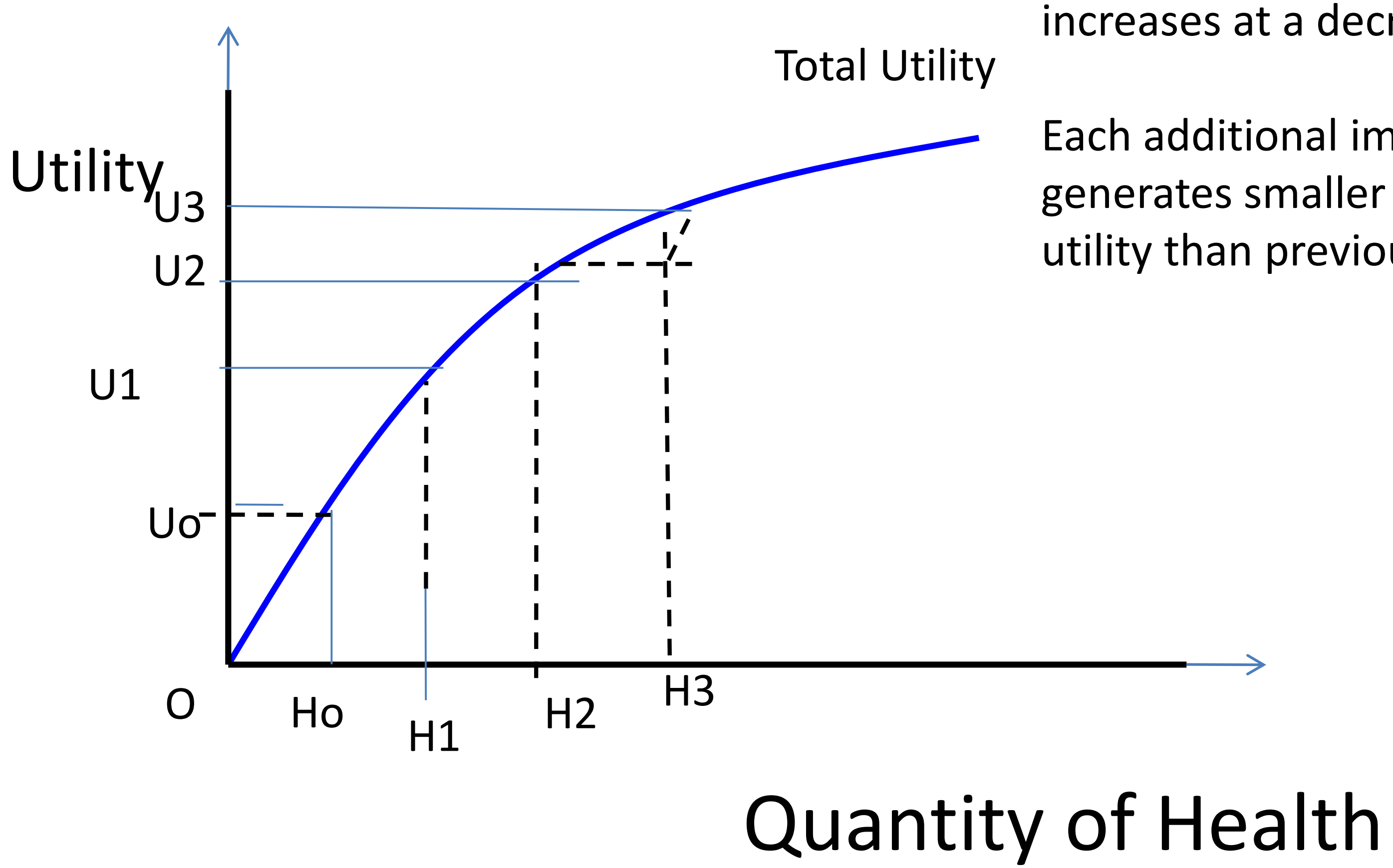




1. TU increases and declines after point q
2. TU increases at decreasing rate
3. Marginal utility declines

Total Utility Curve for Health

- Relation between total utility and health

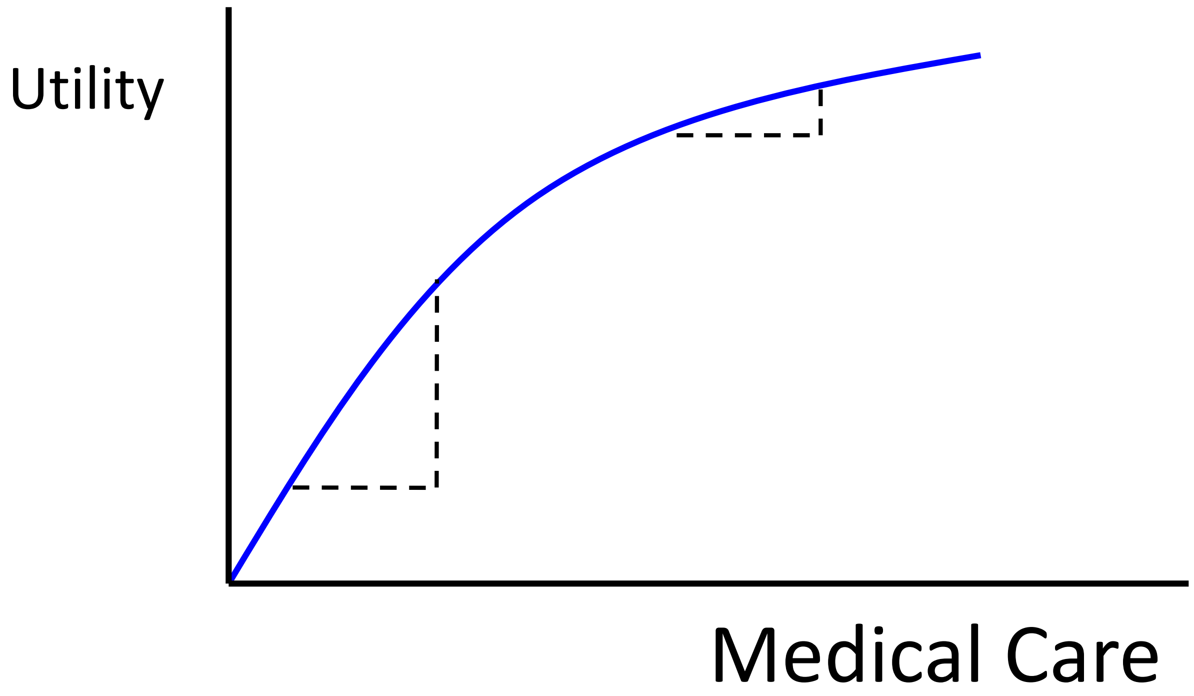


Utility derived from health increases at a decreasing rate

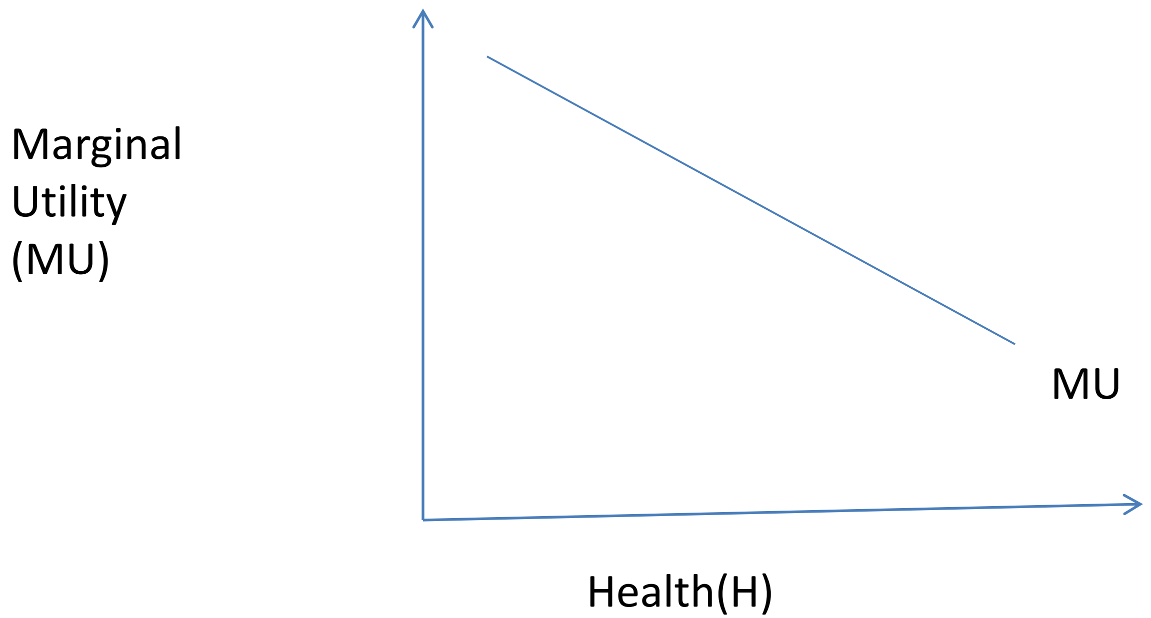
Each additional improvement generates smaller increase in utility than previous one

Medical Care and Utility

We can generally graph the relation between medical care and utility as follows:



Draw a Marginal Utility Curve for Health



$MU_H = \Delta U / \Delta H$

Where MUH is the marginal utility of the last Δ is the change of utility

MU for Medical Care

Marginal utility of medical care **decreases because of two reasons**

(1) Each successive unit of medical care generate utility a smaller improvement in health than the previous one

(2) Each increase in health generate a smaller increase in utility

Numerical example: TP, MP and AP

Units of Labour (L)	Total Product (Q)	Average Product (AP= Q/L)	Marginal Product (ΔQ/ΔL)
1	80	80	80
2	170	85	90
3	270	90	100
4	368	92	98
5	430	86	62
6	480	80	50
7	504	72	24
8	504	63	0
9	495	55	-9
10	470	47	-25

Total Product Curve

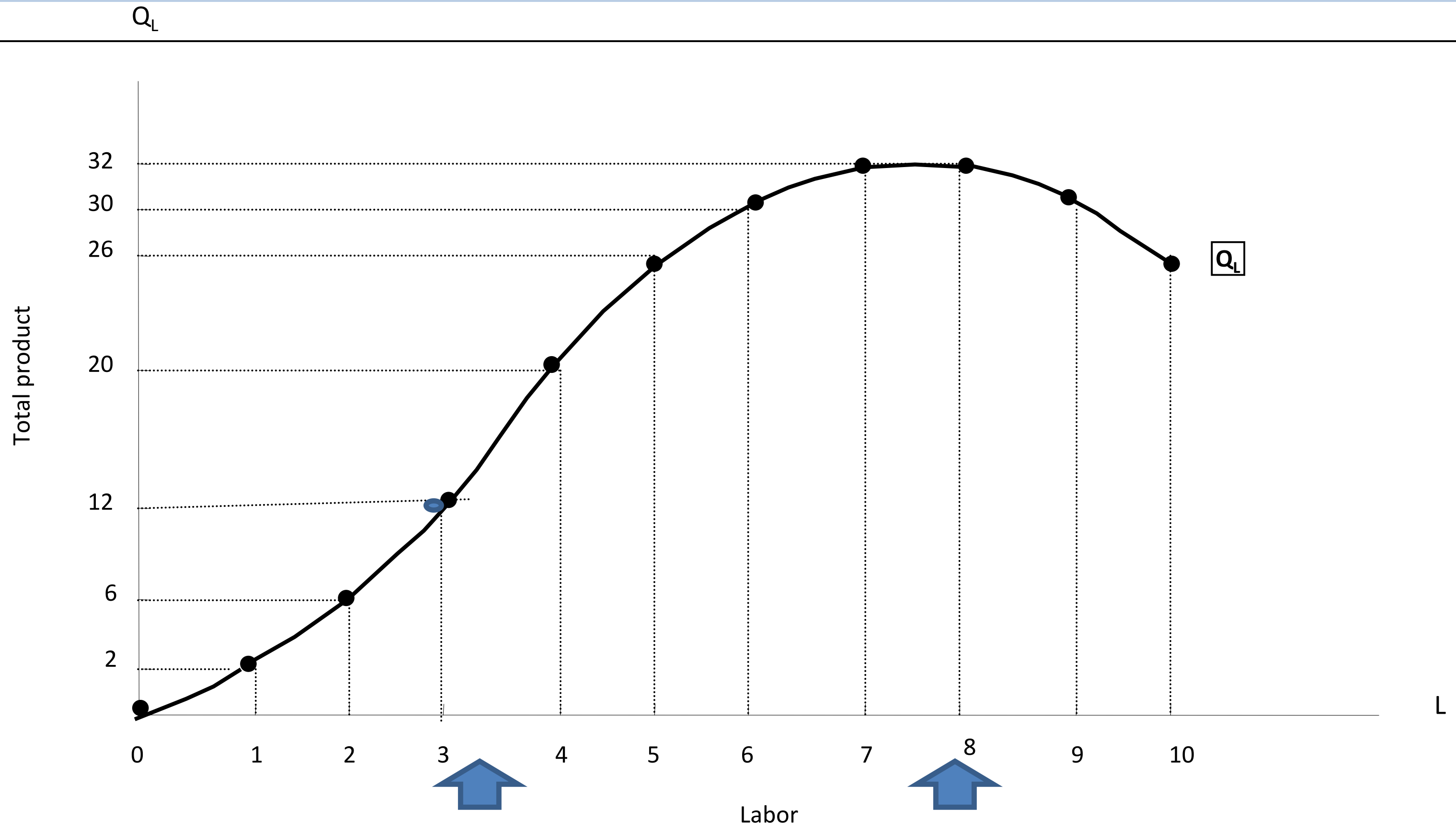


FIGURE Total product curve. The total product curve shows the behavior of total product vis-a-vis an input (e.g., labor) used in production assuming a certain technological level.

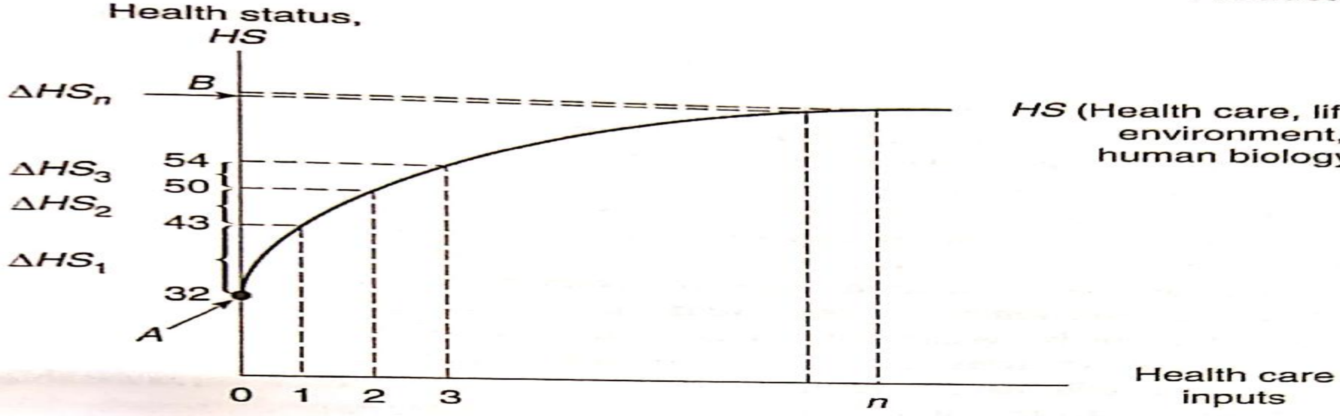
Mathematical Properties

Production Function of Health and Marginal Product of Health Care

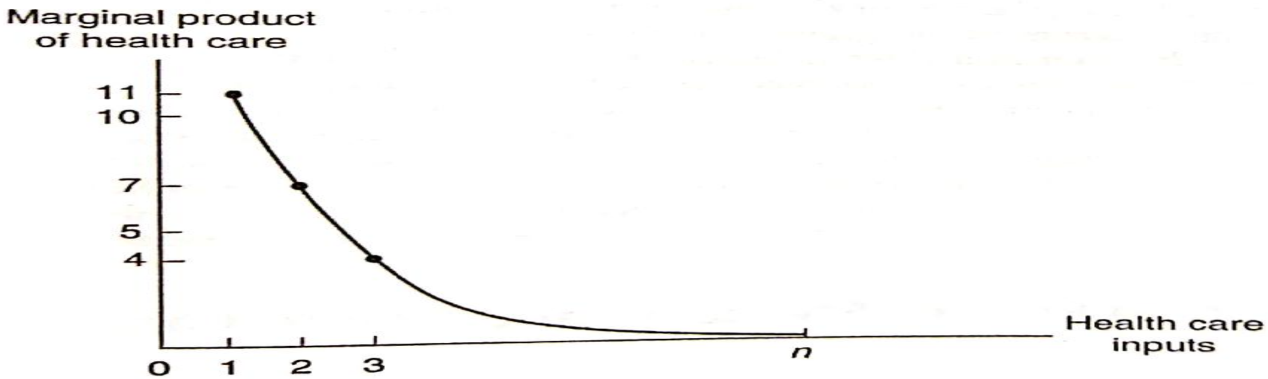
- Medical care is an input in producing health

→ Subject to **law of diminishing marginal productivity**

Production of Health



A. Production Function of Health



B. Marginal Product of Health Care

Theory of Demand

D_x= f (p_x, p_y, I, T) where

D_x: Demand for good X

p_x : Price of good X

p_y : Price of good Y

I: Income of the consumer

T: Taste of the consumer

In short run P_y, I and T are constant .

Dx= f (p_x)……………………………………………………………(1)

Demand for Medical Care

- Medical care assumed to be a **normal good**
- To derive the demand curve for medical care, we must establish the relation between the **quantity of medical services and utility**.
- Health can be viewed as a **durable good** that generates utility.

Demand for medical care is a derived Demand

- Medical care is an input in production of health

- Utility received from consuming health care is **not generated** from health care but from improvement in health

- Health is consumed and produced by individuals

Law of Demand

We know

- ❑ Law of Demand
- ❑ Why Demand Curve Slopes downward?

Factors affecting Demand for Medical Care

Demand for medical care =

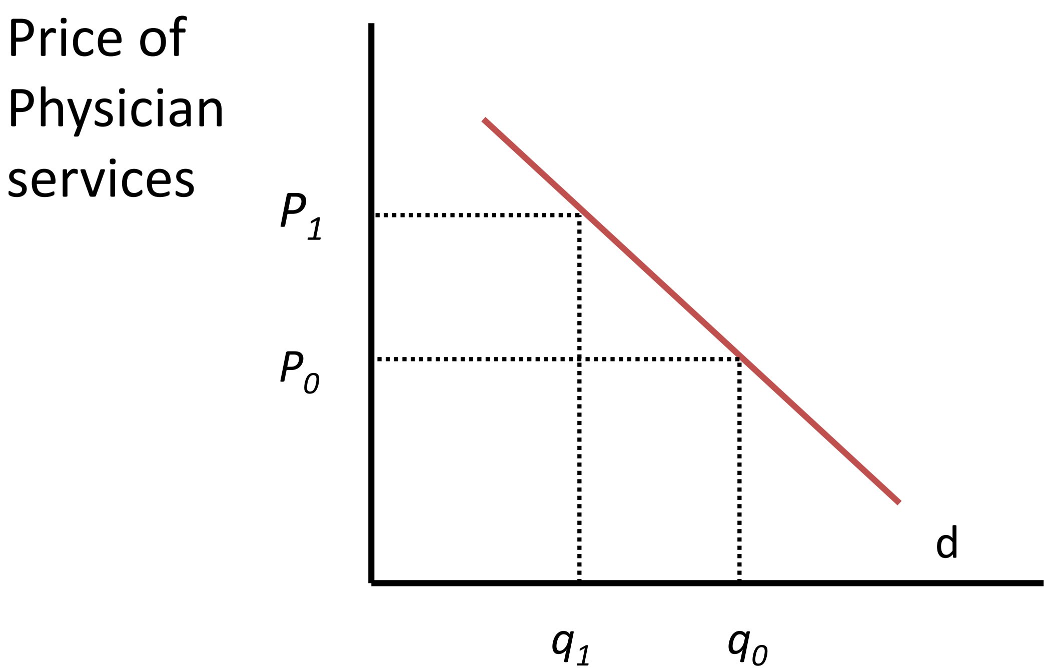
f(Economic factors, Non- economic factors)

Economic Factors	Non-economic factors
Price Income Time Price of substitutes Price of compliments	Sex Marital status Education Life style State of health Quality of care Taste Preferences

1.Demand and Price for Physician Services

- Downward sloping demand curve for physician visits

1.Price of physician visit



Inverse relationship of price and quantity demanded of physician services, ceteris paribus

- Price changes lead to movements *along* D curve

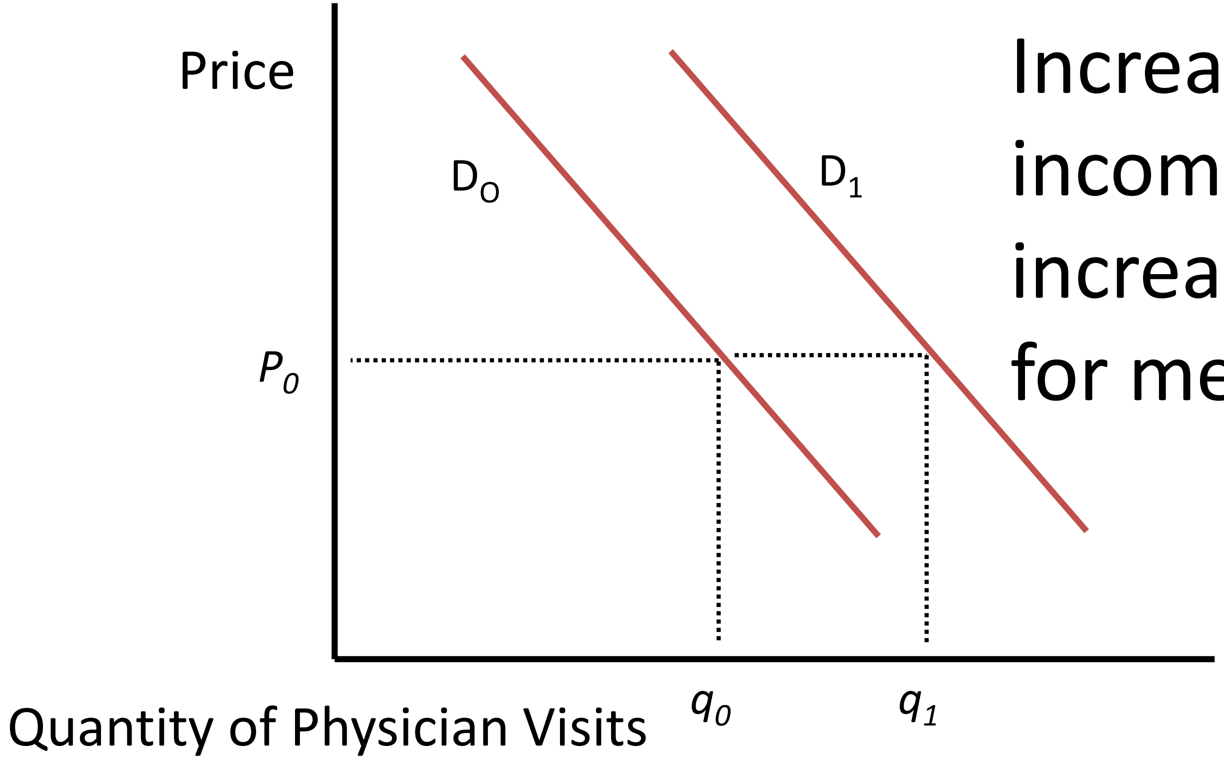
Quiz: Estimate Demand Function

Given the regression estimates of demand equation of
$Qx=1000-3.3P_x+0.001Y$
Where Y is income, P is price

What is the percentage change in demand if price decreases from \$200 to 100\$ with
Given Y=10,000?
$Q1=1000-3.3*200+0.001*10000=1010-660=350$
$Q2=1000-3.3*100+0.001*10000=1010-330=680$
Increase in $QD=100*(330/350)=90\%$

2. Change in Real Income

Real Income



Increase in real income will lead to increase in demand for medical services.

3. Time Cost

Time cost = Monetary **travel cost** + opportunity cost

The opportunity **cost of time** is directly related to a person’s wage rate

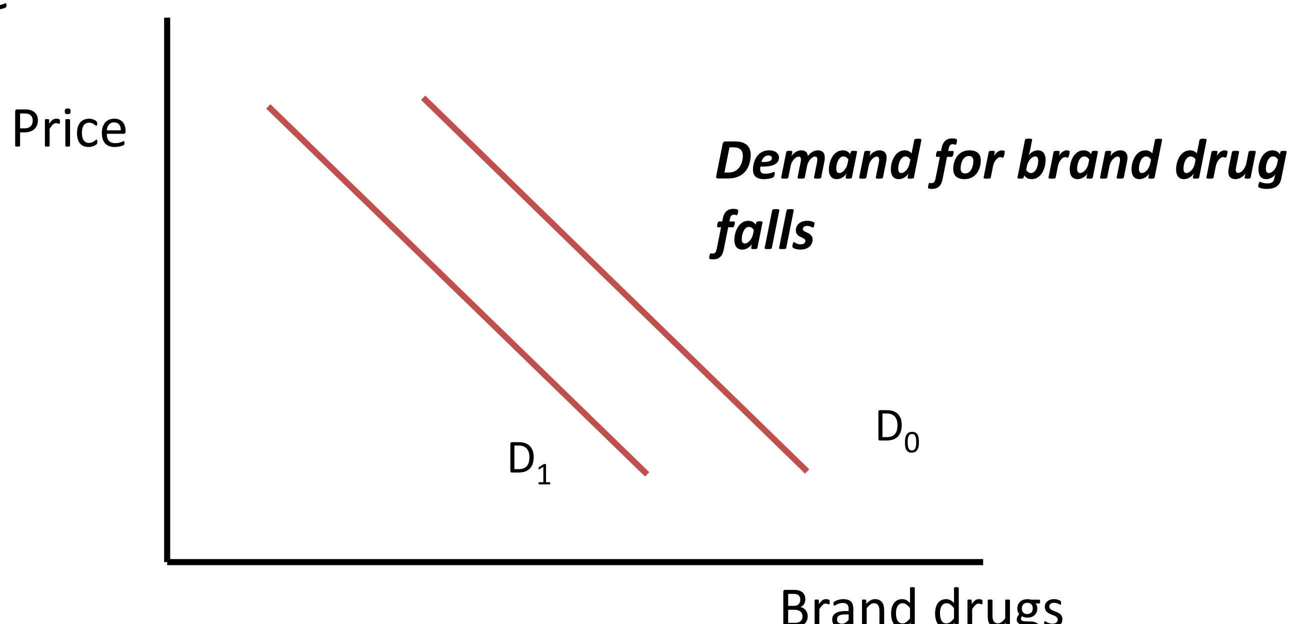
Substitutes

1. Generic and brand drugs
2. Eyeglasses and contact lenses
3. Inpatient services and outpatient services

4. Substitutes

**Substitutes - ** other goods which satisfy the same wants

- e.g. generic and brand name drugs
- If price of brand drug increases, demand for generic drugs increase



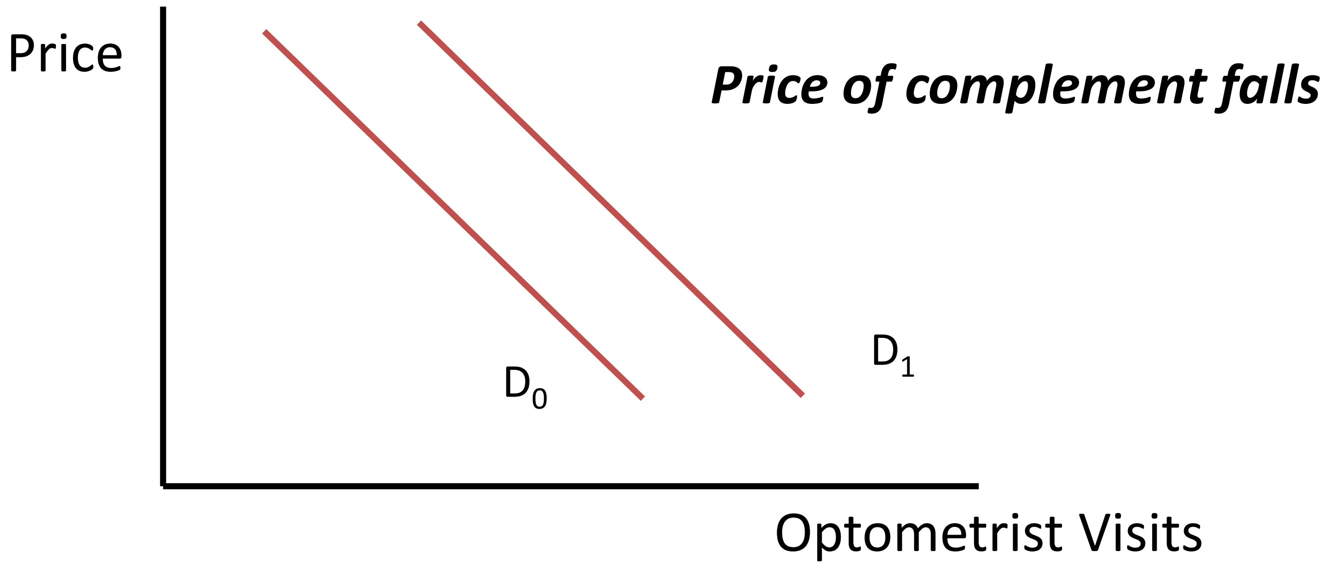
5. Compliments

Complements - Goods which are consumed together

- Contact lenses and optometrist visits
- Obstetric and pediatric services
- Alcohol and cigarettes?

Compliments

If contact lenses become cheaper, demand for optometrist visits increase



Estimating Demand for Medical Care

- Demand for Medical care =
- *f(Economic factors, Non- economic factors…)*

  - Profile
  - state of health
  - quality of care
  - tastes and preferences

Profile and Demand for Health Care

Profile: Age, sex, race/ ethnicity

Age: Use of medical service is an increasing function of age

Female demand more health care services than male because of childbearing

Osteoporosis, mental disorders and Alzheimer's are more prevalent in women than men

Marital status and Demand for Medical Services

Married individual is likely to use medical care, particularly hospital care because availability of spouse to care for him at home

Education and Demand for Medical Services

Relationship of education and use of health care **is difficult to predict**.
Why morbidity is higher in Kerala?

An individual with high level of education may make more **efficient use of health care services** and demand less medical care

Lifestyle and Health

Smoking tobacco and Drinking Alcohol

An individual may compensate for the detrimental health impact of smoking by **consuming more health care services** that translate into increase demand for medical care

State of Health and Demand for Medical Care

State of Health

People with prior history of disease

Severity of Illness

Fuzzy Demand Curve

- We assumed **precise or exact relation** between price and quantity demanded

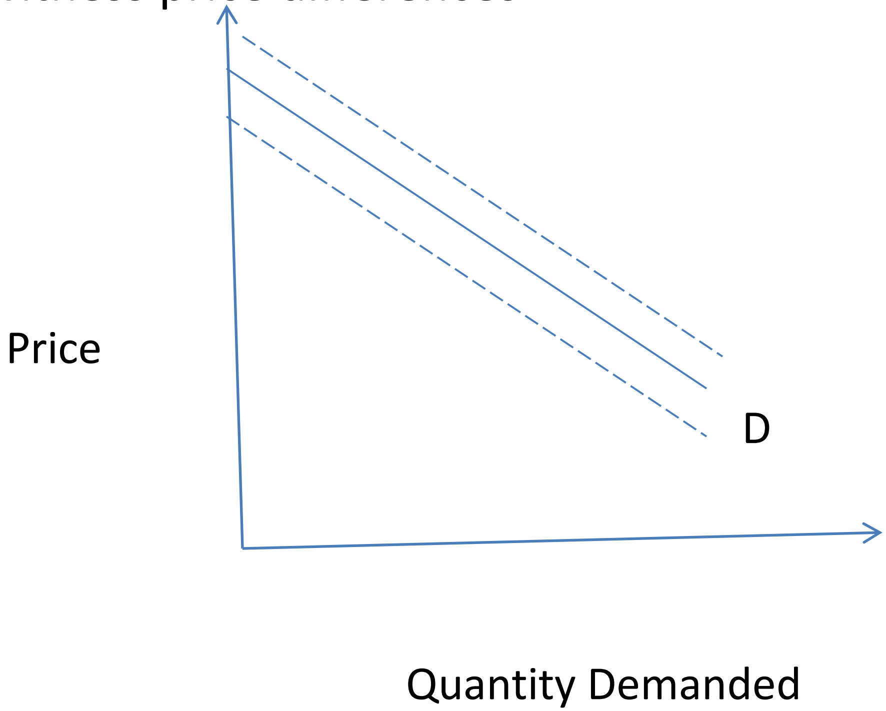
- But the relation between medical care and health **is far from exact????**

- There is considerable lack of medical knowledge concerning certain medical intervention.

- **Physician rather than the consumer choose medical services** which makes the demand curve fuzzier

Fuzzy Demand Curve

1. For a given price, we observe some variation in the quantity or type of medical services rendered
2. For a given quantity or type of medical services we are likely to witness price differences



Possible Fuzziness of the demand for medical services **given uncertainty** and the role of the physicians

Elasticity of Demand for Health Care

Elasticity of Demand for Health Care

- Elasticity measures focus on the r**elative magnitudes of changes** rather than the absolute.

- Health care is a widely **heterogeneous** product.

- Estimation of elasticity is an **empirical question** in health economics

Elasticity

Elasticity of demand is a measure of the responsiveness of demand to changes in one of these determinants.

Why to **measure** elasticity?

Type of Elasticity

A. Price elasticity of demand
B. Income elasticity of demand
C. Cross-elasticity of demand

Price elasticity of Demand for Health care

E_p= percentage change in quantity of health care demanded

....

percentage change in price of health care

$0 \le ep \le \infty$

If $0 < e_p < 1$ => Demand is *inelastic*

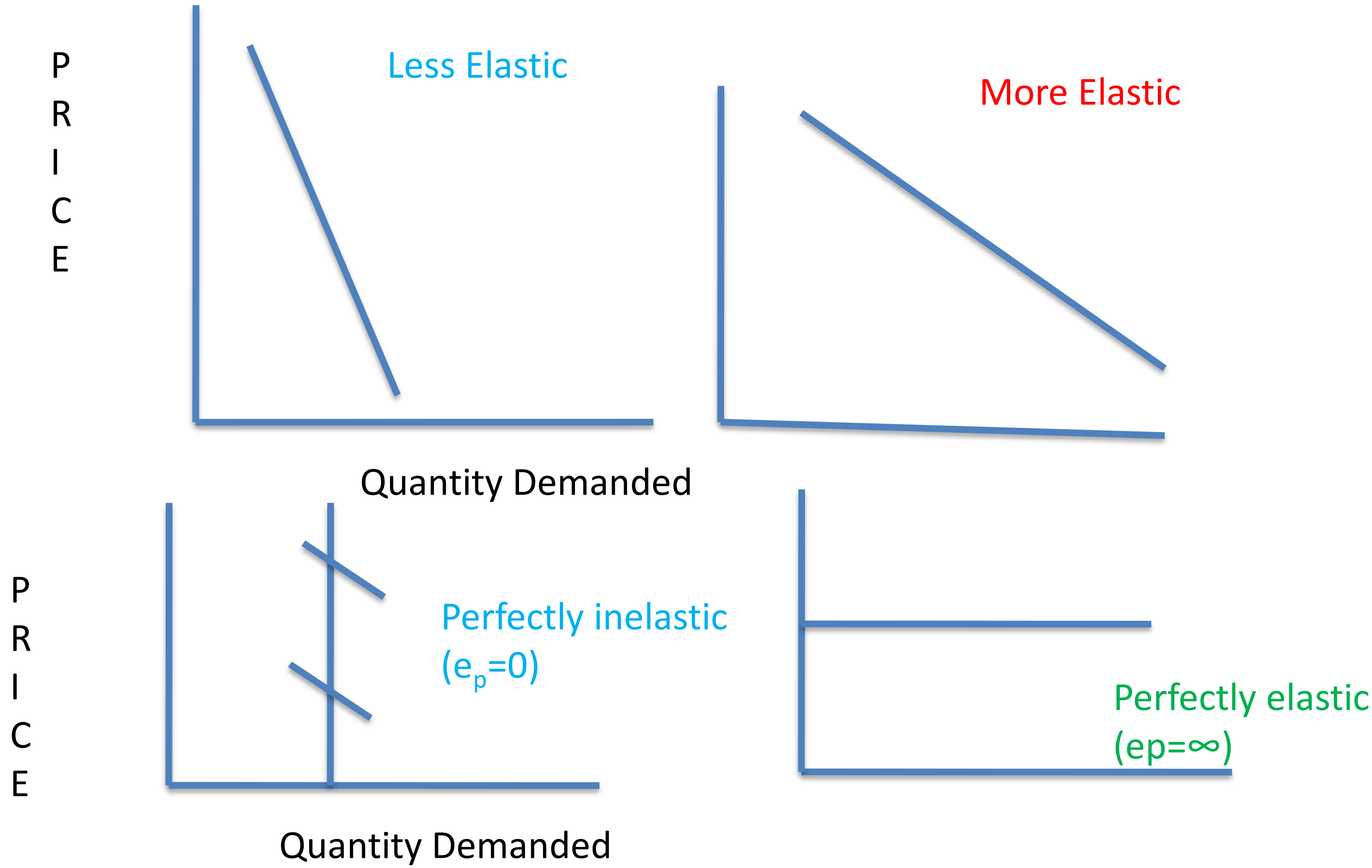
If $e_p = 1$ => Demand is unitary elastic

If $1 < e_p < \infty$ => Demand is ***elastic***

If $e_p = \infty$ => Demand is perfectly elastic

If $e_p = 0$ => Demand is perfectly inelastic

Graphical Presentation



Price elasticity of Demand

Ex 1: A 10% decline in price of doctors' fees resulted 5% change in quantity demanded of services

e_p = 5%/10%= 0.5, Demand is inelastic

Income Elasticity of Demand

e_i= Percentage change in quantity demanded for health care

....

Percentage change in Income

Ex: Increase in consumer income by 10% and quantity demand by 5% , ei= 5/10=0.5

Necessity and Luxury: Income Elasticity

Products that are necessities are expected to be relatively income inelastic

Ex: Treatment for a heart attack is necessities

Cosmetic surgery is luxuries.

Certainly, many types of services fall somewhere in between.

Cross Elasticity of Demand

e_{xy}= Percentage change in quantity demanded for good x

....

Percentage change in price of good y

Ex: Increase in price of brand drug by 5% lead to increase in quantity demanded of generic drug by 4% , exy= 4/5=0.8

Measures used in Elasticity

	Price Measure	Quantity Measure
1	Per capita Income	Health expenditure
2	Consumption Expenditure	Health expenditure
3	Household Income	Household health expenditure
4	Per capita Income	Out-patient visit
5	Co-insurance	Health care visit

Consideration in estimating Elasticity

1. Types of health Care

2. Price of Health Care

3. Time Prices

4. Adverse selection: Person with poor health tend to choose insurance

5. Moral hazard

6. Providers behaviour

Estimation and Interpretation of Demand for Health Services

$$\text{Log } Q = 0.885 - 0.744 \log(P) - 0.50 \log(INC) + 0.253 \log(ADV) - 0.30 \log(PHYSP)$$

(5.52) (4.92) (1.40) (6.64) (0.99)

Adj. $R^2 = 0.30$

N = 243

Where *Q*=Annual dosages demanded of cough and cold medicines

P= Price per dosage of cough and cold medicines

INC=Average income of buyers

ADV= Advertising expenditures on cough and cold medicines

PHYSP=Market price of a physician visit *t*-statistics shown in parentheses below the estimated coefficient

All variables expressed in logarithms so the coefficient estimates can be interpreted as elasticities

Gross man model: A model for the Demand for Health

Dewar DM ,Essentials of Health Economics, Ch 3, The Demand for Health

Grossman, M (1972). On the concept of health capital and the demand for health, ***Journal of Political Economy***, 80:223-255

Utility Maximization

- Let a consumer consume two goods x and y.
- Let p_x = price of good x, and
- P_y =price of good y
- Let M be the money income.
- Let he purchase x quantity of good x and y quantity of good y.
- $p_x . x + p_y . y = M$ ……………….(1)
- Let $U(x, y) = x.y$ ………………..(2)

Contd.....

- Maximize U(x, y) such that $p_x. x + p_y. y = M$
- $L = xy + \lambda(M - p_x. x - p_y. y)$
- $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$ ..... (3)
- $\frac{\partial L}{\partial x} = y - \lambda P_x = 0 \quad \Rightarrow \lambda = \frac{y}{p_x}$ ..... (4)
- $\frac{\partial L}{\partial y} = x - \lambda P_y = 0 \quad \Rightarrow \lambda = \frac{x}{p_y}$ ..... (5)

Contd……

$$\frac{y}{p_x} = \frac{x}{p_y} \Rightarrow \frac{p_x}{p_y} = \frac{x}{y} = MRSx_{,y}$$

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}$$

From Equation (1),

M= p_x. x + p_y. y

= p_x.x + p_x. x = 2p_x.x

$\Rightarrow$ **x** = $\frac{M}{2px}$

Example:

If $p_x = 5$, $p_y = 10$ and $M = 100$

Then, $\mathbf{x} = \frac{100}{2 \times 5} = 10$,

$\mathbf{y} = \frac{100}{2 \times 10} = 5$

And $U = 5 \times 10 = 50$

Two way causation of Income and Health

Health =f (Income)

Income=f(Health)

- How does Income affects health?
- How does health affects income?

Two way causation of Income and Health

- Income affects health through improved nutrition, better health care and improved sanitation

- Health affects income through increasing productivity and healthy days

Effect of Ill health on Income

- ❑ Early retirement
- ❑ Decreasing productivity
- ❑ Increasing dependency burden
- ❑ Increased medical expenditures

Investment in health can stimulate development

Health and Utility: Consumption Perspective

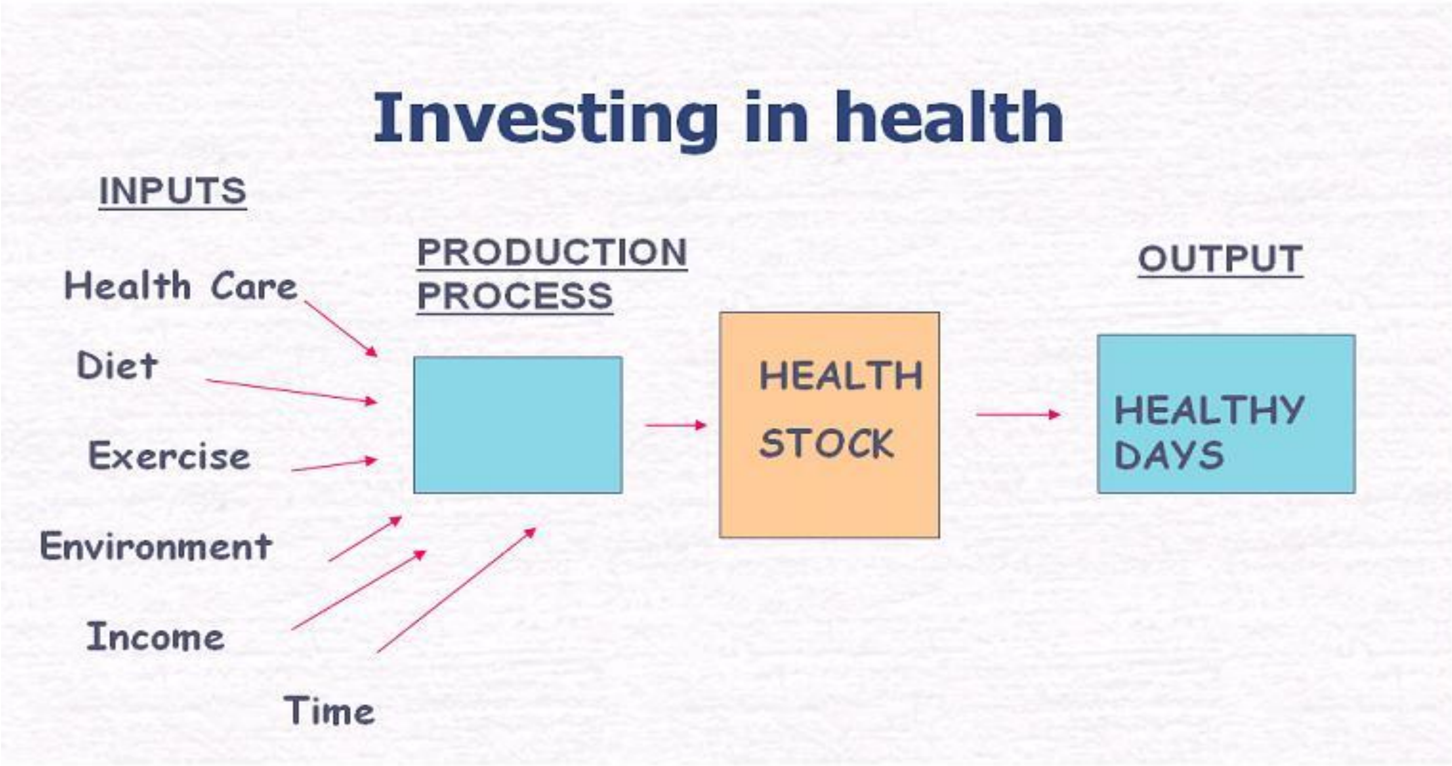
An individual desired to remain healthy because he or she receives utility from an overall improvement in **health.**

Sick days are **unpleasant** and yield **disutility**.

Health and Utility: Investment Perspective

- Good health implies **more healthy days to work** and increase income.
- An increase in stock of health, **reduce time loss for market and non-market activities.**
- Monetary value of time saved is an index of **return to investment** in health

Investing in Health



Gross man model: A model for the Demand for Health

- Proposed by **Michael Grossman in 1970**
- **Grossman Model is an economic model of health**
- **It explains why people demand health and how they allocate money and time to improve or maintain health**

Grossman Model

- In Grossman's model health is both demanded and produced by individuals.

- Health modeled as an **Output** and medical care as **input** in production of health

- As an output, health is demanded because it is a source of utility and ill health reduces happiness and ability to earn.
- Time for health affects total **time available** for production of income and wealth

Grossman Model

Health is demanded because of
- ❑ Work
- ❑ Earn Income
- ❑ Pleasure (Enjoy Life)/ Quality of Life

People **Invest** in health through
a) Medical Care
b) Exercise
c) Diet
to improve health or slow depreciate

Health affect
a) Productivity (work more and earn more)
b) Utility (feel better)

Health As a form of Human Capital

- Wealth is a stock of financial capital because it yields a stream of income

- Health is a stock of human capital that yield a **stream of healthy days.**

- A good health is essential for remaining productive, earning income and for the sake of good quality of life

Health is similar to Capital Stock

- Economist view health as **a durable good or type of capital that provides utility**

- Each person is endowed with a **given stock of health** at the beginning of the period

- The flow of services produced from the stock of capital is consumed continuously over an **individual’s lifetime**

Health As a form of Human Capital

- Grossman model has been influential in health economics.

- Grossman viewed the demand for health as consumption and investment decision

- **Health care is an input into the production of capital good**- the stock of health

Assumption

- Individual inherit an initial stock of health that depreciates with age and can be augmented by investment in health

Model Structure

a)Utility Function

b)Health Production Function

c)Budget Constraint

d)Time Constraint

e)Optimal Choice

a)Utility Function

The theory of health demand starts by assuming that people derive utility from two goods; health (H) and composite of all other commodities (C) over life time.

$U = U(H_t, C_t)$----------------(1)

Determine H* and C*

a)Utility Function

H_t=H(M_t, T_H , E_t) -------------------(2)

C_t= C(X_t, T_c, E_t)--------------------(3)

where M_t is medical care

E_t is the level of education

X is all other good

T_H is the time spent in the production of health

T_c is the time spent in the production of other goods

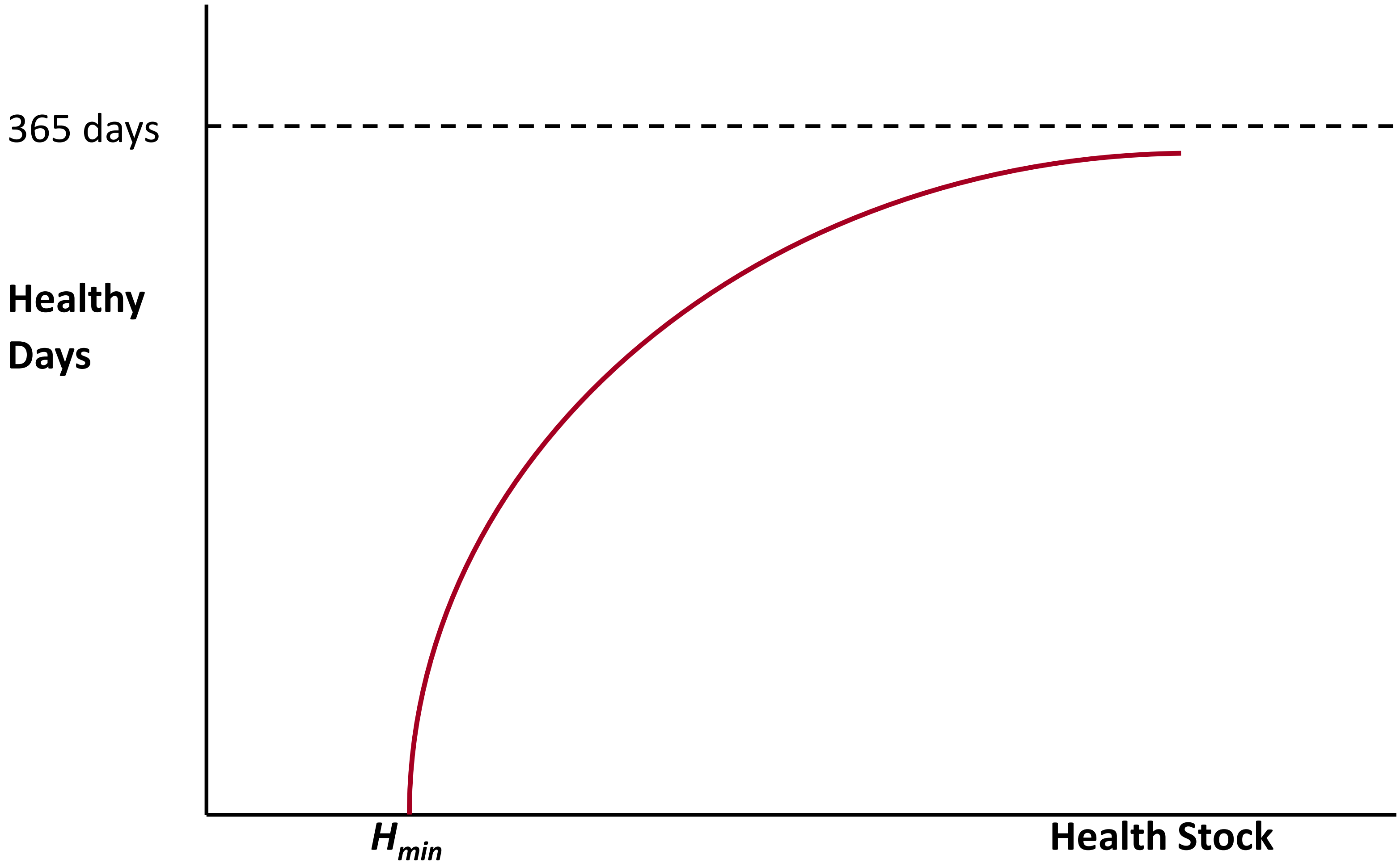
C_t is consumption of non-health goods/commodities

b) Health Production Function

- The stock of health **depreciates** with age and may be augmented by **investment** in health

- Death occurs when an individual stock of health falls below a **critical minimum level**

Relationship of healthy days to Health stock



Production of Healthy Days

- Health is a productive good which produces healthy days
- Greater health stock leads to more healthy days – however with diminishing returns
- H_{min} is health stock minimum – production of healthy days at this point is zero i.e. death
- Natural Maximum of 365 days

b) Health Production Function

- Initial stock of health and rate of depreciation varies from individual to individual

- It depends on many factors, some of which are **uncontrolable**

Capital Stock & Depreciation

$I_t = K_t - K_{t-1}$

Where I_t is Investment in time period t

K_t is Capital Stock in time period t

K_{t-1} is Capital stock in time period t-1

b) Health Production Function

$H_{t+1}= H_t\ (1- \delta)+I_t$ ------------ (4)

Where H_t is health stock at a time t

H_t is health stock at time t

δ : depreciation of health stock

I_t: investment in health at time t

c) Budget Constraint

Assuming that all income is spent on health and non-health such that

$P_{M*} M + P_{x*} X = T_{w*} W = Y_t$---- (5)

W is wage rate

Pm=Price of medical care

Px= Price of all other good

X= Quantity of all non-health goods and services

Y= Income of the consumer

d) Time Constraint

Time budget is divided to time spent on working (T_w) time spent being sick (T_s) , time spent on consumption of non-health goods (T_C) ad time spent on health (T_H)care and so

$T_t = T_{wt} + T_{st} + T_{ct} + T_{Ht}$ ----- (6)

The constraint on expenditure budget is income which depend on how much time is spent on working and the wage rate

e) Optimal Choice

- How much medical care to buy?
  How much non-medical goods to consume?
  Maximise utility subject to
  a)Budget Constraint
  b)Time Constraint
  c) Health Production Function

Optimal Choice: Solving the Grossman Problem

- Maximize
- $\sum_t \delta^t U(H_t, C_t)$ such that

$$W_t = P_c C_t + P_M M_t \text{ and } H_t = (1+\delta)H_{t-1} + f(M_t)$$

- Solution is that Marginal Cost of Investing in Health Capital equals Marginal Benefit of Investing in Health Capital
- Here is all you need to know
  - $r + \delta = W^*G(H)/C(H)$

Grossman Model

- **Proposition:** The stock of health determines the **total amount of time** that can be used to produce income and non-income activities
- **Maximization of the utility function** subjected to constraints
- Equilibrium condition :

Marginal benefits of health capital = Marginal cost

Demand for Health Capital (1)

- Two important concepts: a)Cost of Capital and b) MEI
- Cost of Capital (C)

C = Opportunity cost (cost of foregone alternatives i.e. interest rate) + rate at which health depreciates

$$C = r + \delta$$

Demand For Health Capital

- MEI – Rate of return of amount of invested

  If rate of return on capital good is greater (less) than cost of capital, then the good will (not) be purchased.

Capital good will be purchased only up to point where:

Rate of return = Cost of Capital

What Determines Healthy Choices: Grossman Model

Marginal cost (of investing in H) = Marginal benefits

Marginal cost = $r + \delta$, where

r = rate of interest on other investments

δ = rate of depreciation of health

Marginal benefit $= \frac{(W \times G(H))}{C(H)}$

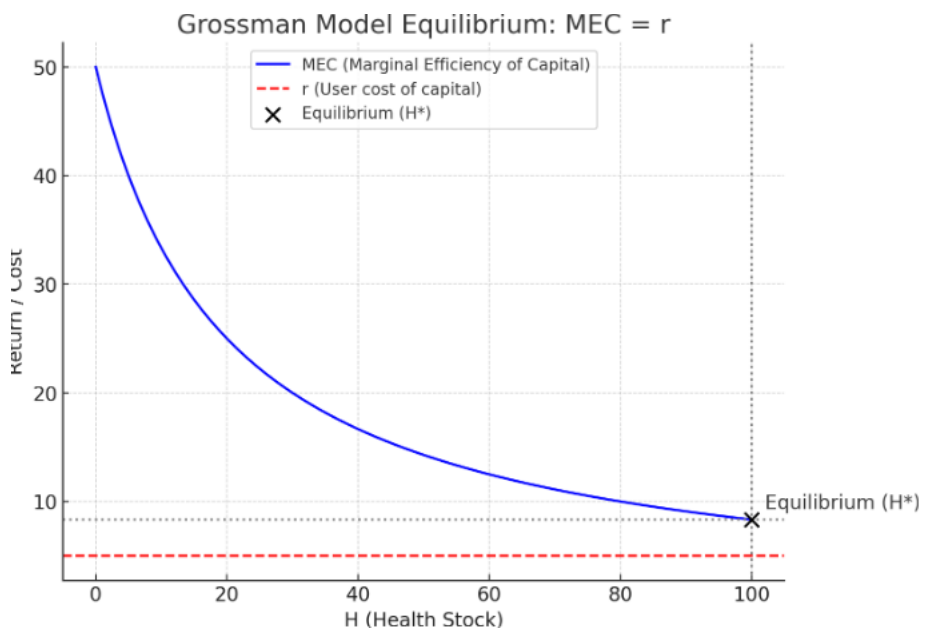
where

W = wage rate,

G(H)= marginal product of health investment

C(H) = direct cost of investment in health

Grossman Model



Grossman Model

To maximise utility, a consumer:

A. Allocate time between work and leisure

B. Spend remaining leisure time on health and non-health activities

C. Spend income earned on health and non-health resources

D. Produce or invest in health capital for future use

Societal and Individual Determinants of Medical Care Utilization in the United States, Andersen and Newman, The Milibank Quarterly, Vol 51, No 1, 1973

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Anderson and Newman Model: Context

- A growing consensus that all people have a right to medical care regardless of their ability to pay for health care

- Certain population groups in USA such as poor, black, Spanish-speaking American and American Indians were not receiving quality and quantity health care

- High expectation that MEDICAL CARE can contribute to the general health level of the population

Societal and Individual Determinants of Medical Care Utilization: Andersen and Newman Model

Was developed in late 1960s

- A theoretical framework for health service utilization that takes both <u>societal and individual</u> determinants
- To understand what are the factors that determine use and non-use of health services?

Andersen and Newman Model

- Individual health care utilization=f(Individual behavior, environment, interaction of individual and societal factors)

- Much attention to individual behavior and less on societal factors.

Societal and Individual Determinants of Medical Care Utilization: Andersen and Newman Model

- Medical Technology and norms

- Health service delivery system

- Individual determinant of utilisation

All these three factors impact health care utilisation

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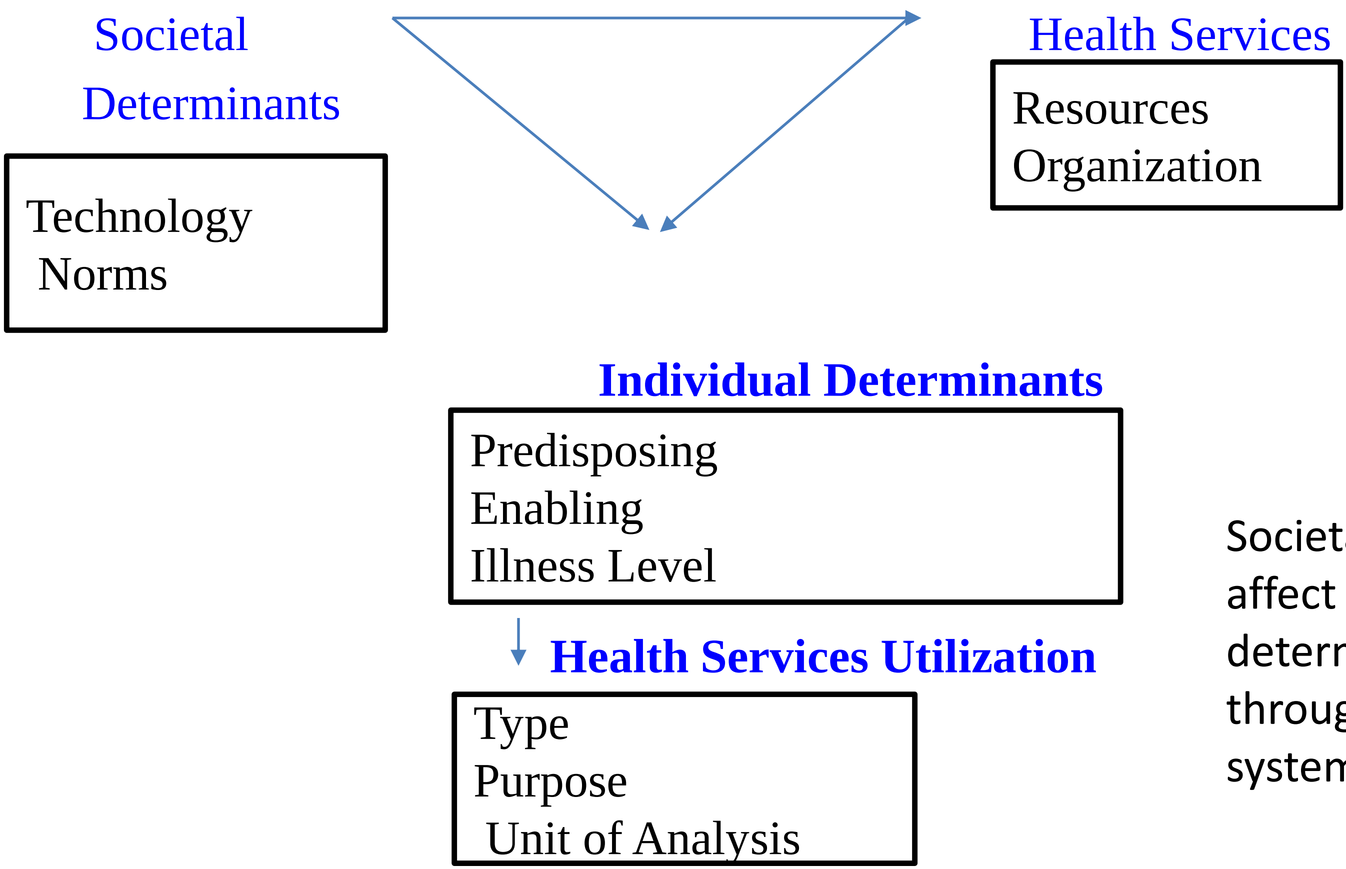
Fig 1. Framework for viewing health services utilization



Societal determinants affect the individual determinants directly or through health service system

Health Services: Societal Determinant

Technology: Medical technology

Decline in mortality due to TB, inflenxa, pnueminia and infections was due to technological advancement.

Mortality due to TB in USA :194 in 1900, 3 by 1969
Development of surgery, radiology and nuclear medicine reduced average length of hospital stay

Norms: Societal control where social system induce compliance on part of the members
Home delivery vs Institutional Delivery
Vaccination

Health Services: Resources

Resources of the system are labour and capital devoted to health care

Health Personnel
Medical Equipment
Materials used for providing health services

Second component of Resources: Geographical Distribution

Health Services: Organisation

Organisation means what the system does with its resources

The components of organization are called access and structure

It refers to the manner in which medical personnel and facilities are coordinated and controlled to provide medical services

Organisation

Access: Means through which the patient gain entry to the medical care system and continue treatment process
Access is low if out-of-pocket payment is high
Access is high if proportion of medical care paid by government, voluntary health insurance or third party payment
Decrease in waiting time

Structure: Characteristics of the system that determine what happen to the patient following entry to the system

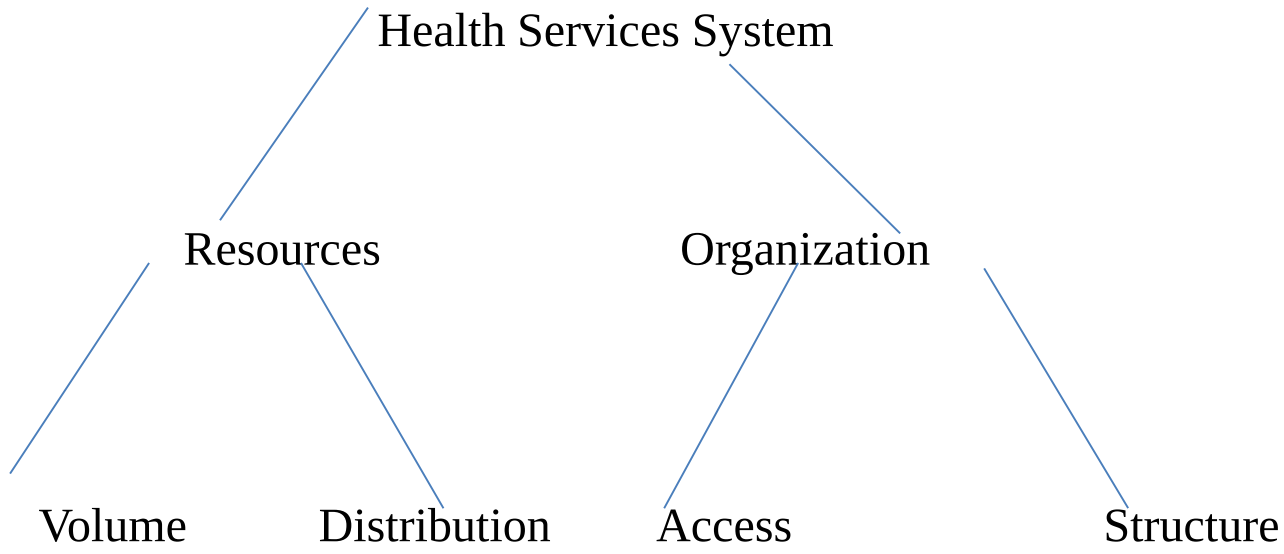
Nature of medical practices who first see the patients
Process of referral
Means of admission to hospital
Characteristics of hospital care

Health Services: Purpose

Primary care: Stopping Illness before it begins
Secondary Care : Process of treatment that return an individual to his previous state of functioning
Tertiary Care : Stabilization for long term irreversible illness.

Custodial Care : Personal needs of the patient but no effort to treat the illness

Fig 3 : The Health Services System



Individual Determinants

1.Predisposing of individual to use services
2. Enabling (ability to secure services) and
3. Illness level (Need)

Fig. Individual determinants of health service utilization

Predisposing

(Susceptible)

Demographic

Age, sex, Marital Status, Past Illness

Social Structure

Education, Occupation, Family Size, Ethnicity, Religion, mobility

Health Beliefs

Value concerning health and illness, attitudes toward health services, knowledge about disease

Enabling

Family

Income, health insurance, third party payment, access to regular sources of care

Community

Ratio of health personnel and facilities to population
Price of health services,
Region of country,
Urban and rural

Need

Perceived

Illness level
Disability,
Symptoms
Diagnoses,
General State

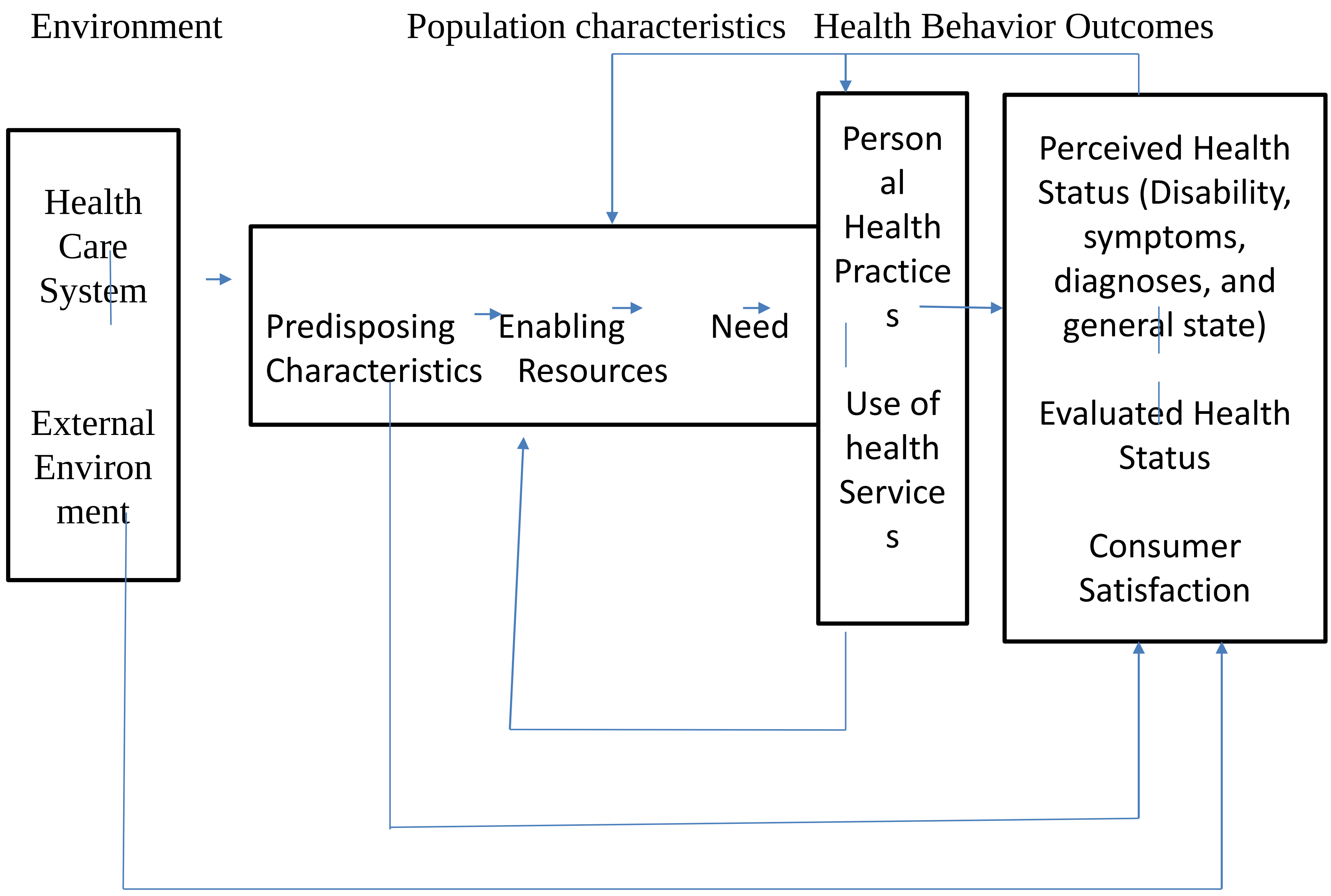
Evaluated

Symptoms

Diagnoses

Use of Health Services

Revised Model



Andersen and Newman Model: Limitations

Autonomy and cognitive impairment as predisposing variables

Organizational factors are not given enough attention

Outcome measures were gross such as health service use and not specific type of services or practioner or episode of illness

External Environment: Political, Physical and Economic
Personal Health Practices: Diet, exercise, self care

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Questions

1. Is medical care a commodity? Discuss the characteristics of medical care as a commodity and how it is different from food. Why demand for health care is termed as derived demand?
2. Draw the demand curve for physician services. Why does the demand for health care is a derived demand?
3. Discuss the economic and non-economic factors that affect the demand for physician services. How does the demand curve shift with respect to change in real income, compliments and substititute .
4. Examine the economic and non-economic factors factors affecting the demand for health care.
5. Define fuzzy demand curve. What are the properties of fuzzy demand curve?. Why demand for medical care is fuzzy?

Questions

5. In your own words, use utility analyses to explain why people demand health? How does the law of diminishing marginal utility fits into this analyses?

6. Discuss the relevance of utility analyses in context of health

7. Medical care is an input in producing health subject to law of diminishing marginal productivity. Substantiate it.

8. Discuss all the factors that affect health function. Explain how a change in each of the following factors would alter the shape of the total product for medical care

   a. An increase in education
   b. An improvement in lifestyle
   c. An improvement in environment

Questions

Q .9. Discuss the two way causation of health and income. Discuss the economist perspective of health. Discuss the salient feature of Gross Man model

Q.10. Define elasticity of demand for health care. What are the consideration needed in estimating elasticity of demand.

Q 11. Discuss in detail the Newman and Anderson Framework in explaining determinant of health care utilization. What were the revision included in revised model?

Q 12. Graph the production of health function HS=10HC0.5E0.3LS0.4HB0.2 in a graph with axes HS and HC. Assuming E=10, LS=5 and HB=7. Graph the marginal product of health inputs. Is it increasing or decreasing? Show how the curve changes when E is increased to 15, where HC is health care, LS is life style, HB is human biology and HS is health.